

A Detailed Discussion for the Proof of Ramanujan's Partition Formula

Jiankun Xu*

Abstract

The partition function $p(n)$, which counts the number of integer partitions of n , has long been a central topic in number theory and combinatorics. In 1918, Hardy and Ramanujan derived a celebrated asymptotic formula for $p(n)$. D. J. Newman subsequently provided an elegant proof of the asymptotic estimate, but many intermediate arguments were omitted, making the proof difficult for readers without an advanced background. This article aims to make the proof more understandable and transparent, offering a detailed discussion of Newman's approach. To this end, the article expands the intermediate steps and illustrates the mathematical techniques involved, such as order estimations, contour integration, and other underlying mathematical methods. In addition, the article briefly reviews the subsequent development of asymptotic estimates for the partition function and emphasizes the significance of Ramanujan's partition formula in related areas.

Keywords: Integer Partition, Partition Function, Ramanujan's Partition Formula, Asymptotic Estimate, Circle Method, Contour integration, Complex Analysis, Combinatorial Analysis, Generating Function, Dominated Convergence Theorem

1 Introduction

A *partition* of a positive integer n is a way to express n as a sum of positive integers, regardless of the order of the summands [1]. For example, the integer 4 has exactly five partitions:

$$4, \quad 3 + 1, \quad 2 + 2, \quad 2 + 1 + 1, \quad 1 + 1 + 1 + 1.$$

The *partition function* $p(n)$ counts the number of such partitions. Thus, $p(4) = 5$ and $p(5) = 7$. It is important to note that order does not matter: $3 + 1$ and $1 + 3$ are considered the same partition.

Although the definition is simple, the behavior of $p(n)$ is surprisingly rich. For small n , $p(n)$ is easy to compute directly, but as n grows, $p(n)$ increases very rapidly:

$$p(10) = 42, \quad p(50) = 204226, \quad p(100) = 190569291.$$

Currently, no simple, straightforward closed-form expression is known for $p(n)$; instead, mathe-

*Jiankun Xu, The Experimental High School Attached to Beijing Normal University, Xicheng District, Beijing, China, 100032. kevinxu703@gmail.com

mathematicians have developed formulas, recurrences, and generating functions to understand its properties [2].

Beyond their intriguing growth behavior, partitions also play a key role in mathematics: they build a bridge between elementary combinatorics and advanced number theory [1]. On the one hand, the partition function $p(n)$ encodes intuitive combinatorial information, making it a fundamental object in combinatorial analysis. On the other hand, its rapid growth reveals nontrivial arithmetic properties that connect to deeper mathematical structures [3].

In addition to pure mathematics, partitions also arise in physics. For example, Zhou and Dai showed that partition functions in statistical mechanics can be expressed in terms of integer partitions [4].

The study of partitions does not have a clearly identifiable origin, but it may date back to Leonhard Euler in the 18th century, when Euler discovered that partitions can be represented by a generating function [5]:

$$\sum_{n=0}^{\infty} p(n)q^n = \prod_{k=1}^{\infty} \frac{1}{1-q^k}, \quad |q| < 1. \quad (\text{eq1})$$

This infinite product not only encodes the sequence $\{p(n)\}$ but also reveals deep connections between partitions and other areas of mathematics, such as modular forms and q -series[3]. In fact, this fundamental representation of integer partitions also shows the basic generating function of $p(n)$, which we will discuss in the next section.

In the 19th century, James Joseph Sylvester contributed to the combinatorial study of integer partitions [6]. He investigated partitions under specific constraints, such as requiring all parts to be distinct or odd only. He provided meaningful insight into how combinatorial methods can be applied to analyze the structure and properties of partitions.

In the field of estimation, a landmark breakthrough occurred in 1918, when G. H. Hardy and S. Ramanujan presented their famous asymptotic formula for $p(n)$ [7, 2].

Theorem 1.1 (Ramanujan's Partition Formula). *For large n , we have*

$$p(n) \sim \frac{\exp\left(\pi\sqrt{\frac{2n}{3}}\right)}{4n\sqrt{3}}.$$

This formula gives an accurate estimate of $p(n)$. This result was one of the earliest applications of the *circle method*, a powerful analytic tool in number theory[7]. In addition, this asymptotic formula was also independently invented by J. V. Uspensky in 1920 [8].

In 1951, D. J. Newman provided a simple and elegant proof of this formula [9], making the core ideas more accessible to a wider audience. Newman later refined and improved his proof in 1962 [10], streamlining the argument while preserving its beauty. However, Newman's articles were not very readable for a person who is not familiar with advanced mathematics.

Thus, the article aims to explain some of the details in Newman's proof of Ramanujan's partition formula [10], including intermediate steps that are often omitted. By carefully expanding on these steps, the article hopes to make the argument much more accessible to readers with a high school

background in mathematics, requiring only basic calculus and complex numbers, so that they can appreciate not only the final result, but also the ideas and techniques that make it work. Also, the article discusses some basic values and applications of Ramanujan's partition formula.

In Section 2, we will show the necessary knowledge, including definitions and theorems, that can help a high-school student understand how the proof of our main goal works, and readers are also able to find related proofs from the cited works. Subsequently, in Section 3, we will discuss the main content of the proof of Ramanujan's partition formula, apply previous formulas step by step, and finally compute the estimate of the partition function. Next, in Section 4, we will show the further development on the estimation of the partition function. Finally, in Section 5, we will briefly provide some relations between partitions and other mathematical topics and some applications of partitions, stressing the importance of Ramanujan's partition formula. In addition, readers may find a concrete comparison between the partition function and Ramanujan's partition formula in the appendix.

2 Technical Preliminaries

Here, we discuss some information that is important to understand this paper. Readers can search for detailed verification in given references.

The proof will begin with the generating function of the partition function, as generating functions always have closely related properties with their original functions. Readers may be unfamiliar with generating functions, so it is crucial to make a brief introduction.

Definition 1 (Generating Function). *Let $\{q_n\}_{n=0}^{\infty}$ be a sequence of real (or complex) numbers. The generating function of this sequence is the formal power series*

$$G(x) := \sum_{n=0}^{\infty} q_n x^n.$$

Usually, we focus only on those generating functions that converge so that we can study their understandable properties.

Remark. *Readers may think of the generating function as a single expression that stores the whole sequence, in which the coefficient of x^n in $G(x)$ is exactly a_n . For example, the sequence $1, 3, 5, \dots$ has the generating function*

$$1 + 3x + 5x^2 + 7x^3 + \dots$$

Note that x is just a placeholder.

Now we can focus on the generating function of the partition function.

Definition 2 (The Generating Function of the Partition Function). *For the partition function $p(n)$ with $p(0) = 1$, the generating function is defined as*

$$f(z) = \sum_{n=0}^{\infty} p(n) z^n = \prod_{m=1}^{\infty} \frac{1}{1 - z^m} \quad (|z| < 1).$$

Remark. *We now explain how this expression works. First, note that considering the geometric*

series, we have

$$\prod_{m=1}^{\infty} \frac{1}{1 - z^m} = \prod_{m=1}^{\infty} (1 + z^m + z^{2m} + \dots) = \prod_{m=1}^{\infty} \sum_{n=0}^{\infty} z^{mn}.$$

Then choose several terms from each factor in the expression. e.g., we have

$$\begin{aligned} z^8 &= z^{1 \cdot 2} \cdot z^{2 \cdot 1} \cdot z^{3 \cdot 0} \cdot z^{4 \cdot 1} \cdot z^{5 \cdot 0} \cdot z^{6 \cdot 0} \dots \\ &= z^{1 \cdot 0} \cdot z^{2 \cdot 1} \cdot z^{3 \cdot 2} \cdot z^{4 \cdot 0} \cdot z^{5 \cdot 0} \cdot z^{6 \cdot 0} \dots \end{aligned}$$

We can establish a mapping of these to the partitions $8 = 1 + 1 + 2 + 4$ and $8 = 2 + 3 + 3$. Likewise, it is possible to establish a bijective (mutual) mapping between every single partition and every combination of different powers of z . In this way, every distinct expression contributes to the coefficients in the series function. Thus, the coefficients can represent the numerical values of partitions.

Definition 3 (Argument of Complex Number). For a nonzero complex number $z \in \mathbb{C}$, its argument is any real number θ such that

$$z = |z|e^{i\theta}.$$

We denote such θ by $\arg(z)$, defined modulo 2π .

We now define the Taylor expansion, which provides estimates of functions, although it appears in most calculus textbooks.

Theorem 2.1 (Taylor Expansion). Let f be a function with continuous derivatives up to order $n + 1$ on an open interval containing a and x . Then there exists ξ between a and x such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - a)^{n+1},$$

where the first term is called the n -th order Taylor polynomial of f at a , and the last term is the error term.

Furthermore, if f is infinitely differentiable at a and the remaining term tends to 0 as $n \rightarrow \infty$ for x in some neighborhood of a (equivalently, the series converges to $f(x)$ there), then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n,$$

which is called the Taylor expansion of f about a .

Proof. The proof of the Taylor expansion is well-known. Readers can find a rigorous proof in most common textbooks [11]. $\square$

Remark. For x within the radius of convergence, we can use the Taylor polynomial to estimate a function.

Then we show an identical relation that connects an infinite series with a finite number.

Theorem 2.2 (Basel Problem). *The infinite series*

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges, and its sum is given by

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Proof. The rigorous proof is well-known, but it uses knowledge beyond our discussion. Therefore, we encourage readers to learn from articles themselves [12, 13]. $\square$

We do not discuss the proof of the exact Basel problem because it is too sophisticated and advanced, but we show a necessary corollary in our proof here.

Corollary 2.3 (Rough Basel Problem).

$$\sum_{n=2}^{\infty} \frac{1}{n^2} < 1.$$

Proof. The above theorem directly leads to this result. Still, we prefer to show a clear proof. Since for all $n \geq 2$, we have $\frac{1}{n^2} < \frac{1}{n(n-1)}$, we get:

$$\sum_{n=2}^{\infty} \frac{1}{n^2} < \sum_{n=2}^{\infty} \frac{1}{n(n-1)} = \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right).$$

Therefore, we have

$$\sum_{n=2}^{\infty} \frac{1}{n^2} < \sum_{n=2}^{\infty} \frac{1}{n-1} - \frac{1}{n} = \lim_{n \rightarrow \infty} 1 - \frac{1}{n} = 1.$$

$\square$

We use geometric series when discussing the infinite series in the article.

Theorem 2.4 (Geometric Series).

$$\frac{a}{1-z} = a + az + az^2 + \cdots = \sum_{n=0}^{\infty} az^n$$

holds for all $|z| < 1$.

Proof. Considering the partial sum, we find that the limit converges to $a/(1-z)$. $\square$

We then define the Big-O notation, which will allow us to describe the growth rate of auxiliary functions related to $p(n)$ without tracking every constant. This is essential when we discuss asymptotic estimates.

Definition 4 (Big-O Notation). *Let $f(x)$ and $g(x)$ be functions defined for sufficiently large x . We write*

$$f(x) = O(g(x)) \quad \text{as } x \rightarrow \infty$$

if there exist constants $C > 0$ and $\delta > 0$ such that

$$|f(x)| \leq C|g(x)| \quad \text{as } 0 < |x - a| < \delta.$$

This means that $f(x)$ grows at most as fast as $g(x)$ near $x = a$.

Now we give the definition of the supremum, which helps us make bounded estimates for functions. Readers may understand the supremum as a broader notion of the **maximum**, though it need not be attained.

Definition 5 (Supremum). *Let A be a nonempty subset of $\mathbb{R}$ (or of $\mathbb{C}$, where comparison is based on modulus $|\cdot|$). A number $s \in \mathbb{R}$ is called the **supremum** (least upper bound) of A if:*

- $a \leq s$ for all $a \in A$ (i.e., s is an upper bound), and
- for any $\varepsilon > 0$, there exists $a \in A$ such that $s - \varepsilon < a$.

We write $s = \sup A$.

Now we show the total variation, which gives a direct sense of how intense the oscillation of a function is. This concept will appear later in an important inequality (Lemma 3.2.1).

Definition 6 (Total Variation on an Interval). *Let $f : [a, b] \rightarrow \mathbb{R}$ (or $f : [a, b] \rightarrow \mathbb{C}$). The total variation of f on $[a, b]$ is defined as:*

$$V_a^b(f) = \sup \left\{ \sum_{j=1}^n |f(x_j) - f(x_{j-1})| : a = x_0 < x_1 < \cdots < x_n = b, n \in \mathbb{N} \right\}.$$

That is, $V_a^b(f)$ is the supremum of the total difference in modulus on all partitions of $[a, b]$.

Remark. *In the case of complex-valued functions, the absolute value $|\cdot|$ is interpreted as the modulus. The total variation definition remains valid, since modulus defines a metric and allows for measuring function properties.*

We need complex integrals, so we give the necessary information and explanations here. Readers may learn more about complex integrals in textbooks [13].

Definition 7 (Analytic (Holomorphic) Function). *Let $U \subset \mathbb{C}$ be an open set. A function $f : U \rightarrow \mathbb{C}$ is called analytic (or holomorphic) if it is complex differentiable at every point of U , i.e., for every $z_0 \in U$, the limit*

$$f'(z_0) := \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists.

Remark. *Complex differentiability requires this limit to be the same regardless of the direction from which z approaches z_0 , which is a stronger condition than real differentiability and can fail even if partial derivatives exist. The precise conditions ensuring complex differentiability involve the Cauchy-Riemann equations, which characterize when a function is holomorphic. For brevity, we omit these details here and encourage readers to learn about the Cauchy-Riemann equations [13].*

The rules in complex calculus have similarities to those in real calculus. We discuss some basic mathematical techniques for complex integrals here so that readers can understand how the estimate

of $p(n)$, in terms of complex integrals, works.

Definition 8 (Complex Integral). *Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a piecewise smooth curve in the complex plane, parametrized as*

$$\gamma(t) = x(t) + iy(t), \quad t \in [a, b].$$

For a complex function f , the integral of f along γ is defined by

$$\int_{\gamma} f(z) dz := \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

Basically, a complex integral is computed by parameterizing the curve in the complex plane and then reducing the integral to an ordinary real integral with respect to the parameter t .

Now, with a basic understanding of complex integrals, we discuss some theorems in complex analysis used in our proof. Cauchy's theorem points out a basic conclusion in complex integrals, which helps us connect an integral on complex plane to one on the real axis (Lemma 3.2.2).

Theorem 2.5 (Cauchy's Theorem). *Let $U \subset \mathbb{C}$ be an open set, and suppose $f : U \rightarrow \mathbb{C}$ to be a holomorphic function. Suppose $\gamma : [a, b] \rightarrow U$ is a piecewise smooth, closed curve that is null-homotopic in U . Then*

$$\oint_{\gamma} f(z) dz = 0.$$

Proof. The proof includes knowledge beyond our discussion. Readers can find available details in complex analysis textbooks. [13]. $\square$

Remark. *The symbol “ $\oint$ ” resembles “ $\int$ ” but specifically denotes an integral taken along a closed curve, which is called a contour integral. Readers may consider the word “null-homotopic” as meaning that the curve can be contracted to a simple point without obstacles like zero points.*

A simpler version of Cauchy's theorem assumes that γ is a simple closed curve bounding a region entirely contained in U . In this case, the conclusion

$$\oint_{\gamma} f(z) dz = 0$$

remains valid.

To understand this theorem, readers should note that simple complex integrals are usually computed with some parameterization related to boundary conditions, while the enclosed curve provides a nice condition for computation.

Corollary 2.6 (Coefficient Extraction). *Let $G(z) = \sum_{k=0}^{\infty} q_k z^k$ be a power series with radius of convergence $R > 0$. For any positively oriented simple closed contour γ around the origin with $|\zeta| < R$ on γ , the coefficient q_n can be recovered by the contour integral*

$$q_n = \frac{1}{2\pi i} \oint_{\gamma} \frac{G(\zeta)}{\zeta^{n+1}} d\zeta.$$

Proof. Since the series converges uniformly on $|\zeta| = r < R$, we may interchange the sum and the

integral:

$$\oint_{\gamma} \frac{G(\zeta)}{\zeta^{n+1}} d\zeta = \oint_{\gamma} \sum_{k=0}^{\infty} q_k \frac{\zeta^k}{\zeta^{n+1}} d\zeta = \sum_{k=0}^{\infty} q_k \oint_{\gamma} \zeta^{k-n-1} d\zeta.$$

Now recall the elementary fact [13] that for integers m ,

$$\oint_{\gamma} \zeta^m d\zeta = \begin{cases} 2\pi i, & m = -1, \\ 0, & m \neq -1. \end{cases}$$

Hence, only the term $k = n$ survives, giving

$$\oint_{\gamma} \frac{G(\zeta)}{\zeta^{n+1}} d\zeta = q_n \cdot 2\pi i.$$

□

Remark. *This corollary is crucial in analytic combinatorics: it allows us to translate analytic properties of the generating function into asymptotic information about its coefficients. In particular, we will apply it to the generating function of the partition function, writing*

$$p(n) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z^{n+1}} dz.$$

This identity is the analytic bridge that enables us to derive the Ramanujan's partition formula by estimating contour integrals (Theorem 3.4).

As the partition function has an exponential order, and we will use integrals to estimate $p(n)$, we have to consider integrals of exponents. Integrals similar to $\int_D e^{ax+b} dx$ are well-known, and integrals where the exponent of x is greater than 2 will not be used, so we give an integration formula, the Gaussian integral, which works for integrals where the exponent of x is 2 (Theorem 3.4).

Theorem 2.7 (Gaussian Standard Integral).

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$$

Proof. Let

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx.$$

We write

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy.$$

Changing to polar coordinates with substitution

$$x = r \cos \theta, \quad y = r \sin \theta, \quad \text{and} \quad dx dy = r dr d\theta,$$

we have

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta = 2\pi \int_0^{\infty} r e^{-r^2} dr.$$

Let $u = r^2$, so $du = 2r dr$. Then

$$I^2 = 2\pi \cdot \frac{1}{2} \int_0^\infty e^{-u} du = \pi \cdot [-e^{-u}]_0^\infty = \pi.$$

Thus $I = \sqrt{\pi}$. □

Remark. *The proof requires some basic knowledge of multivariable calculus, so readers may gain a rough understanding of the approach.*

Corollary 2.8 (Gaussian Integral). *For any real number $a > 0$, $b \in \mathbb{R}$, we have*

$$\int_{-\infty}^\infty e^{-at^2+bt} dt = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}}.$$

Proof. By the Gaussian Standard Integral, we have

$$\int_{-\infty}^\infty e^{-at^2+bt} dt = e^{\frac{b^2}{4a}} \int_{-\infty}^\infty e^{-a(t-\frac{b}{2a})^2} dt = e^{\frac{b^2}{4a}} \sqrt{\frac{\pi}{a}}.$$

□

Note that factorials appear in the Taylor expansion of exponents (Theorem 3.4), so we discuss an approximation formula of them.

Theorem 2.9 (Stirling's Approximation Formula). *For large $n \in \mathbb{N}$, the factorial satisfies the asymptotic formula*

$$n! \sim n^n e^{-n} \sqrt{2\pi n}.$$

Proof. By the definition of the Gamma function [13], there is

$$\Gamma(z+1) = \int_0^\infty t^z e^{-t} dt = \int_0^\infty e^{z \ln t - t} dt.$$

Let $t = zx$, then

$$\Gamma(z+1) = z \int_0^\infty e^{z \ln(zx) - zx} dx = z \int_0^\infty e^{z \ln z + z \ln x - zx} dx = z^{z+1} \int_0^\infty e^{z(\ln x - x)} dx.$$

Note that the function $e^{z(\ln x - x)}$ achieves its maximum at $x = 1$. Around this point, we use a second-order Taylor polynomial:

$$\ln x - x \approx -1 - \frac{(x-1)^2}{2} \quad \text{for } x \approx 1.$$

So we find

$$\int_0^\infty e^{z(\ln x - x)} dx \approx e^{-z} \int_0^\infty e^{-\frac{z(x-1)^2}{2}} dx.$$

Next, let $u = \sqrt{\frac{z}{2}}(x-1)$, so $dx = \sqrt{\frac{2}{z}} du$. The integral becomes:

$$e^{-z} \sqrt{\frac{2}{z}} \int_{-\infty}^\infty e^{-u^2} du = e^{-z} \sqrt{\frac{2\pi}{z}}.$$

Thus,

$$\Gamma(z+1) \sim z^{z+1} \cdot e^{-z} \cdot \sqrt{\frac{2\pi}{z}} = \left(\frac{z}{e}\right)^z \sqrt{2\pi z}.$$

Taking $z = n \in \mathbb{N}$, we conclude that

$$n! \sim n^n e^{-n} \sqrt{2\pi n}.$$

□

When we try to use integrals to estimate a function related to the partition function $p(n)$, we may consider what will happen as $n \rightarrow \infty$. However, it is not always proper to exchange the positions of the integral sign and the limit sign, or

$$\lim_{k \rightarrow a} \int_D f(x, k) dx \neq \int_D \lim_{k \rightarrow a} f(x, k) dx,$$

because as a parameter in the function approaches infinity, the integrated function may change its properties. For example, we consider the function

$$f_n(x) = \begin{cases} n, & 0 < x < \frac{1}{n}, \\ 0, & \frac{1}{n} \leq x \leq 1. \end{cases}$$

We have

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = 1 \neq \int_0^1 \lim_{n \rightarrow \infty} f_n(x) dx = 0.$$

Therefore, it is necessary to find out when this approach can be used. The dominated convergence theorem explains this topic. It appears in the integral estimate related to $p(n)$ (Theorem 3.4).

Theorem 2.10 (Dominated Convergence Theorem). *Let $(X, \mathcal{M}, \mu)$ be a measure space, and let $\{f_n\}_{n=1}^\infty$ be a sequence of measurable functions $f_n : X \rightarrow \mathbb{R}$.*

Suppose:

1. $f_n(x) \rightarrow f(x)$ for almost every $x \in X$;
2. There exists an integrable function $g : X \rightarrow [0, \infty)$ such that $|f_n(x)| \leq g(x)$ for all n and almost every x .

Then we have

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu,$$

and in particular f is integrable.

Remark. *The dominated convergence theorem (DCT) is a fundamental result in analysis, which provides sufficient conditions under which the order of limit and integration can be interchanged.*

Readers may understand the theorem in this way: if a sequence of “common” functions converges pointwise to a limit, and all of them are bounded in magnitude by an “integrable” function, then the integrals also converge to the integral of the limit function.

For readers unfamiliar with Lebesgue integration, the statement regarding “measure space” and

“measurable function” can be viewed as a generalization of similar results in Riemann integration but with broader applicability.

With DCT, we can deduce a corollary that helps us determine the convergence of functions (Claim 3.4.2).

Corollary 2.11 (Derivatives of Gaussian Integral). *For $a > 0$, $b, c \in \mathbb{R}$, and integer $n \geq 0$,*

$$\frac{d^n}{db^n} \int_{-\infty}^{\infty} e^{-ax^2+bx+c} dx = \int_{-\infty}^{\infty} x^n e^{-ax^2+bx+c} dx = k_n(a, b) e^{\frac{b^2}{4a}+c},$$

where $k_n(a, b)$ depends on a and b .

Proof. We interpret the n -th derivative with respect to b as a limit of difference quotients. This allows us to justify, via the dominated convergence theorem, the exchange of the derivative and the integral.

Although we do not explicitly construct a dominating function, the rapid decay of $x^n e^{-ax^2+bx+c}$ as $x \rightarrow \infty$ ensures that such an integrable function exists and the conditions of the theorem are satisfied.

Thus, from Theorem 2.8 (the Gaussian integral), differentiating under the integral sign yields

$$\frac{d^n}{db^n} \int_{-\infty}^{\infty} e^{-ax^2+bx+c} dx = \int_{-\infty}^{\infty} x^n e^{-ax^2+bx+c} dx = \frac{d^n}{db^n} e^c \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}} = k_n(a, b) e^{\frac{b^2}{4a}+c}.$$

□

3 Proof of Ramanujan’s Partition Formula

This section basically derives from Newman’s proof [10], providing rigorous and specific details that help readers to quickly understand the proof of Ramanujan’s partition formula.

First, we estimate the order of the generating function of the partition function $f(z)$. Next, we set up a special function $q(n)$, finding another estimate of $f(z)$. Subsequently, we apply the two estimates in a contour integral (called the circle method) in order to evaluate the error between $p(n)$ and $q(n)$. Finally, we evaluate $q(n)$.

Order Evaluation

We start with proving Theorem 3.1.

Theorem 3.1 (Order of $|f(z)|$). *The generating function of integer partitions $f(z)$ satisfies the following:*

$$|f(z)| < \exp \left(\frac{1}{|1-z|} + \frac{1}{1-|z|} \right).$$

The order evaluation of $|f(z)|$ later works in the complex integral that helps estimate the error of Ramanujan’s partition formula.

Proof. Taking the logarithms for the definition of $f(z)$, expanding the logarithms with Theorem 2.1 (the Taylor expansion), and subsequently using the geometric series, we have:

$$\log f(z) = - \sum_{n=1}^{\infty} \log(1 - z^n) = \sum_{m,n=1}^{\infty} \frac{z^{mn}}{n} = \sum_{n=1}^{\infty} \frac{z^n}{n(1 - z^n)}. \quad (\text{eq2})$$

With the triangle inequality $|z_1 + z_2| \leq |z_1| + |z_2|$, we have

$$\left| \sum_{n=1}^{\infty} \frac{z^n}{n(1 - z^n)} \right| \leq \frac{|z|}{|1 - z|} + \sum_{n=2}^{\infty} \frac{|z^n|}{n|1 - z^n|}.$$

Then we use a trick that amplifies the second term through the geometric series:

$$\begin{aligned} \frac{|z|}{|1 - z|} + \sum_{n=2}^{\infty} \frac{|z^n|}{n|1 - z^n|} &< \frac{1}{|1 - z|} + \sum_{n=2}^{\infty} \frac{n|z^n|}{n^2(1 - |z^n|)} \\ &= \frac{1}{|1 - z|} + \sum_{n=2}^{\infty} \frac{1}{n^2} \cdot \frac{n|z^n|}{(1 - |z|)(|z|^{n-1} + |z|^{n-2} + \dots + 1)} \\ &= \frac{1}{|1 - z|} + \frac{1}{1 - |z|} \sum_{n=2}^{\infty} \frac{1}{n^2} \frac{n}{|z|^{-1} + |z|^{-2} + \dots + |z|^{-n}} \\ &< \frac{1}{|1 - z|} + \frac{1}{1 - |z|} \sum_{n=2}^{\infty} \frac{1}{n^2}, \end{aligned}$$

where we use $|z| < 1$. After that, we can use Corollary 2.3 to deduce that

$$\frac{1}{|1 - z|} + \frac{1}{1 - |z|} \sum_{n=2}^{\infty} \frac{1}{n^2} < \frac{1}{|1 - z|} + \frac{1}{1 - |z|}$$

Finally, taking the exponent on both sides, we complete the proof. □

Establishing Connection with a Computable Function

Now we try to find a numerical estimate of $f(z)$.

Theorem 3.2. Setting $\phi(z) = \left(\frac{1-z}{2\pi}\right)^{1/2} \exp\left[\frac{\pi^2}{12} \left(-1 + \frac{2}{1-z}\right)\right]$, we obtain

$$f(z) = \phi(z) [1 + O(1 - |z|)] \quad (|z| < 1, |1 - z| < 2(1 - |z|)).$$

Remark. The limitation $|z| < 1, |1 - z| < 2(1 - |z|)$ reflects a certain area on the complex plane. It will work later in the complex integrals.

Now, we give a mathematical technique regarding the total variation from G. Pólya and G. Szegő [14].

Lemma 3.2.1 (Total Variation Theorem). Let $g : [0, \infty) \rightarrow \mathbb{R}$ be integrable on every finite interval

and assume $V(g)$ converges. Fix $t > 0$ and, for an integer $N \geq 1$, set the finite partial sum

$$S_N(t) := t \sum_{n=1}^N g(nt).$$

Then for every $N \geq 1$,

$$\left| S_N(t) - \int_0^{Nt} g(u) du \right| \leq t V_0^{Nt}(g).$$

Consequently, if the improper integral $\int_0^\infty g(u) du$ converges and the sequence $S_N(t)$ converges as $N \rightarrow \infty$, then

$$\left| t \sum_{n=1}^\infty g(nt) - \int_0^\infty g(u) du \right| \leq tV.$$

Proof. Partition $[0, Nt]$ into the small intervals $I_n = ((n-1)t, nt]$, $n = 1, \dots, N$. Then we have

$$\int_0^{Nt} g(u) du = \sum_{n=1}^N \int_{I_n} g(u) du.$$

For each n ,

$$\int_{I_n} g(u) du - t g(nt) = \int_{I_n} (g(u) - g(nt)) du,$$

hence

$$\left| \int_{I_n} (g(u) - g(nt)) du \right| \leq \int_{I_n} \sup_{u \in I_n} |g(u) - g(nt)| du \leq t V_{I_n}(g).$$

Summing over n gives

$$\left| \int_0^{Nt} g(u) du - t \sum_{n=1}^N g(nt) \right| \leq t \sum_{n=1}^N V_{I_n}(g) = t V_0^{Nt}(g) \leq tV,$$

which proves the finite- N estimate. Finally, as $N \rightarrow \infty$, we complete the proof. $\square$

Remark. By rotation, the same inequality applies to any ray L from the origin on the complex plane. For $w \in L$ and g of the bounded variation on L , we have:

$$\left| w \sum_{n=1}^\infty g(nw) - \int_L g(u) du \right| \leq |w| V_L(g).$$

We omit the rigorous proof of this proposition, which can be roughly understood by replacing the length of the real values with the moduli of the complex values.

Lemma 3.2.2. Let $\Theta = (\theta_0, \theta_1) \subset [0, 2\pi]$ be an open interval of angles and define the area

$$S = \{re^{i\theta} | r > 0, \theta \in \Theta\}.$$

Suppose f satisfies the following conditions

1. f is analytic in S ;

2. There exist constants $M > 0$, $R_0 > 0$ and $\varepsilon > 0$ such that

$$|f(re^{i\theta})| \leq \frac{M}{r^{1+\varepsilon}} \quad \text{for all } r \geq R_0, \theta \in [\theta_0, \theta_1];$$

3. The improper integral of f near the origin along any ray in S converges.

Then for any two angles $\alpha, \beta \in \Theta$,

$$\int_0^{\infty e^{i\alpha}} f(z) dz = \int_0^{\infty e^{i\beta}} f(z) dz.$$

Proof. WLOG, suppose $\alpha < \beta$. For $R > R_0$, consider the contour C_R consisting of:

- L_α : the ray from 0 to $Re^{i\alpha}$;
- Arc_R : the circular arc $\{Re^{i\theta} : \theta \in [\alpha, \beta]\}$ oriented counterclockwise;
- L_β : the ray from $Re^{i\beta}$ back to 0.

By Theorem 2.5 (Cauchy's theorem), $\oint_{C_R} f(z) dz = 0$, so

$$\int_{L_\alpha} f(z) dz + \int_{Arc_R} f(z) dz + \int_{L_\beta} f(z) dz = 0.$$

On Arc_R , $|dz| = R d\theta$ and $|f(z)| \leq MR^{-1-\varepsilon}$, hence as $R \rightarrow \infty$

$$\left| \int_{Arc_R} f(z) dz \right| \leq M(\beta - \alpha)R^{-\varepsilon} \rightarrow 0.$$

Letting $R \rightarrow \infty$ gives

$$\int_{L_\alpha} f(z) dz = - \int_{L_\beta} f(z) dz,$$

or

$$\int_0^{\infty e^{i\alpha}} f(z) dz = \int_0^{\infty e^{i\beta}} f(z) dz.$$

For $\alpha > \beta$, the proof is similar. □

With these lemmas, we can prove Theorem 3.2.

Proof of Theorem 3.2. Set $z = e^{-w}$, where $w = a + bi$ is a complex number with $|\text{Im}(w)| \leq \pi$. Under the circumstance $|z| < 1$, $|1 - z| < 2(1 - |z|)$, w satisfies

$$1 - e^{-a} = |1| - |z| \leq |1 - z| < 2(1 - e^{-a}) \quad \Rightarrow a > 0,$$

which means $\text{Re}(w) > 0$, and there exists a K such that $|\arg w| < K < \pi/2$.

Since the domain of the generating function $f(z)$ is $|z| < 1$, setting $z = e^{-w}$ can transform the complex variable z into a complex variable w with $\text{Re}(w) > 0$, which means the infinite series

summation can be simplified by integrating along the real axis. This is the core technique in Newman's proof.

In terms of w , the identity (eq2) becomes

$$\log f(z) = \sum_{n=1}^{\infty} \frac{z^n}{n(1-z^n)} = \sum_{n=1}^{\infty} \frac{e^{-nw}}{n(1-e^{-nw})} = \sum_{n=1}^{\infty} \frac{1}{n(e^{nw} - 1)}.$$

We add and subtract $\frac{1}{2} \log(1 - e^{-w})$ and $\frac{\pi^2}{6w}$ on the right-hand side, and use Theorem 2.2 (the Basel problem) and Theorem 2.1 (the Taylor expansion) of logarithms:

$$\log f(z) = \frac{\pi^2}{6w} + \frac{1}{2} \log(1 - e^{-w}) + w \sum_{n=1}^{\infty} \left(\frac{1}{nw(e^{nw} - 1)} - \frac{1}{n^2 w^2} + \frac{e^{-nw}}{2nw} \right). \quad (\text{eq3})$$

Now note that the sum in the last term of (eq3) is related to nw . Thus, consider the function

$$g(u) = \frac{1}{u(e^u - 1)} - \frac{1}{u^2} + \frac{e^{-u}}{2u}$$

and the ray $L = \{u = r e^{i \arg w} \mid r > 0\}$ since all points in terms of nw are on this ray.

From Newman [9], we know

$$\int_0^{\infty} g(u) du = -\frac{1}{2} \log(2\pi).$$

It is clear that $g(u)$ satisfies all conditions in Lemma 3.2.2, so we can apply this theorem with the real axis and the ray L to find out that

$$\int_0^{\infty} g(u) du = \int_L g(u) du = -\frac{1}{2} \log(2\pi).$$

Since $g(u)$ is differentiable and $g'(u)$ is integrable, according to the definition of integrals, the total variation $V_L(g)$ can be rewritten by:

$$\begin{aligned} V_L(g) &= \sup \left\{ \lim_{r \rightarrow \infty} \sum_{j=1}^n |f(x_j) - f(x_{j-1})| : 0 = |x_0| < |x_1| < \dots < |x_n| = r, x_k \in L \right\} \\ &= \int_L |g'(u)| |du|. \end{aligned} \quad (\text{eq4})$$

Remark. We can have a rough estimate of $V(g)$ on $u > 0$ to find that it is a finite number, which helps us to understand the properties of $V_L(g)$, although it is not rigorous: on $u > 0$, $g(u)$ is monotonically increasing (the exponential term controls the order) and always less than 0; according to the Taylor expansion, we also have

$$\lim_{u \rightarrow 0^+} g(u) = -\frac{5}{12}.$$

We estimate the order of $|g'(u)|$ to deduce that $V_L(g)$ is bounded. To begin with, we have the

differentiation of $g(u)$:

$$g'(u) = -\frac{(e^u - 1) + ue^u}{u^2(e^u - 1)^2} + \frac{2}{u^3} + \frac{-e^{-u}}{2u} - \frac{e^{-u}}{2u^2}.$$

We may evaluate $|g'(u)|$ for $|\arg u| < K$: when $|u| < 1$, since $g(u)$ is analytic in $|u| < 2\pi$, there exists a constant M'' such that $|g'(u)| \leq M''$; when $|u|$ is large, by analyzing the asymptotic behavior of each term, we find that there exists a positive constant M' such that:

$$|g'(u)| \leq \frac{M'}{|u|^3},$$

since the exponential terms control the functions as $u \rightarrow \infty$. Thus, the total variation V_L of $g(u)$ on the ray L is bounded to be a finite constant provided $|\arg w| < K$.

Using the above results, we can use Lemma 3.2.1 to get that

$$w \sum_{n=1}^{\infty} g(nw) = -\frac{1}{2} \log 2\pi + O(w). \quad (\text{eq5})$$

Then we go back to (eq3). Note that

$$\begin{aligned} z = e^{-w} &\sim 1 - w \quad \text{and} \quad 1 - z \sim w - \frac{w^2}{2} = w \left(1 - \frac{w}{2}\right) \\ \Rightarrow \frac{1}{1 - z} &= \frac{1}{w} \cdot \frac{1}{1 - \frac{w}{2}} \sim \frac{1}{w} \left(1 + \frac{w}{2}\right) = \frac{1}{w} + \frac{1}{2}, \end{aligned}$$

so by (eq5), we deduce that

$$\begin{aligned} \log f(z) &= \frac{\pi^2}{6w} + \frac{1}{2} \log(1 - e^{-w}) + w \sum_{n=1}^{\infty} g(nw) \\ &= \frac{\pi^2}{6w} + \frac{1}{2} \log(1 - e^{-w}) - \frac{1}{2} \log 2\pi + O(w) \\ &= \frac{\pi^2}{6w} + \frac{1}{2} \log \left(\frac{1 - e^{-w}}{2\pi} \right) + O(w) \\ &\sim \frac{\pi^2}{6(1 - z)} - \frac{\pi^2}{12} + \frac{1}{2} \log \frac{1 - z}{2\pi} + O(1 - z) \\ &\sim \log \phi(z) + O(1 - |z|). \end{aligned} \quad (\text{eq6})$$

Note that the limitation of z implies that $1 - z \sim 1 - |z|$. Finally, taking exponents from both sides completes the proof. $\square$

Estimation of the Error

In this part, we prove Theorem 3.3.

Theorem 3.3. Set $\phi(z) = \sum q(n)z^n$, then

$$p(n) = q(n) + O(n^{-5/4} \exp(\pi\sqrt{2n/3})).$$

This will simplify the problem to the computation of the coefficients of the elementary function $\phi(z)$. Before we dive into the detailed proof, it is necessary to discuss the circle method, the main idea of the proof of theorem 3.3, in advance.

Circle Method

The circle method, developed by Hardy and Ramanujan in 1918 [7], is originally an analytic technique in number theory that studies generating functions by integrating them over the unit circle in the complex plane. By decomposing the integration path into major and minor arcs, one can isolate the main contributions and estimate the error terms. In other words, when we need to compute an integral in terms of

$$\oint_{\gamma} F(z) dz$$

where γ is a simple closed curve on the complex plane, we can decompose γ into two specific parts γ_1 and $\gamma_2 = \gamma \setminus \gamma_1$ so that

$$\oint_{\gamma} F(z) dz = \int_{\gamma_1} F(z) dz + \int_{\gamma_2} F(z) dz,$$

and the contributions from the integrals on the two curves are distinctly different.

Nowadays, this method has been widely used to study partition functions and other additive problems in number theory [15].

Now we turn back to the proof.

Proof of Theorem 3.3. By Corollary 2.6,

$$p(n) - q(n) = \frac{1}{2\pi i} \oint_C \frac{f(z) - \phi(z)}{z^{n+1}} dz,$$

where C is any simple closed curve in the complex plane. From Newman [10], the curve is chosen as the circle $C = \left\{ z : |z| = 1 - \frac{\pi}{\sqrt{6n}} \right\}$. We divide C into

$$A = \left\{ z \in C : |1 - z| < \frac{\pi\sqrt{2}}{\sqrt{3n}} = 2(1 - |z|) \right\}$$

and $B = C \setminus A$, as shown in Figure 1. The choice and the division of the curve provide a practical way to establish a numerical relation between the estimate of the complex integrals and the final evaluation.

We first compute the integration on B . The length of B is $O(1)$. By Theorem 3.1, we obtain the estimate:

$$\begin{aligned} \int_B \frac{f(z) - \phi(z)}{z^{n+1}} dz &= O \left(\int_B \frac{|f(z)| + |\phi(z)|}{z^n} |dz| \right) \\ &= O \left(\int_B |z|^{-n} \left\{ \exp \left(\frac{\pi^2}{6|1-z|} \right) + \exp \left(|1-z|^{-1} + (1-|z|)^{-1} \right) \right\} |dz| \right). \end{aligned}$$

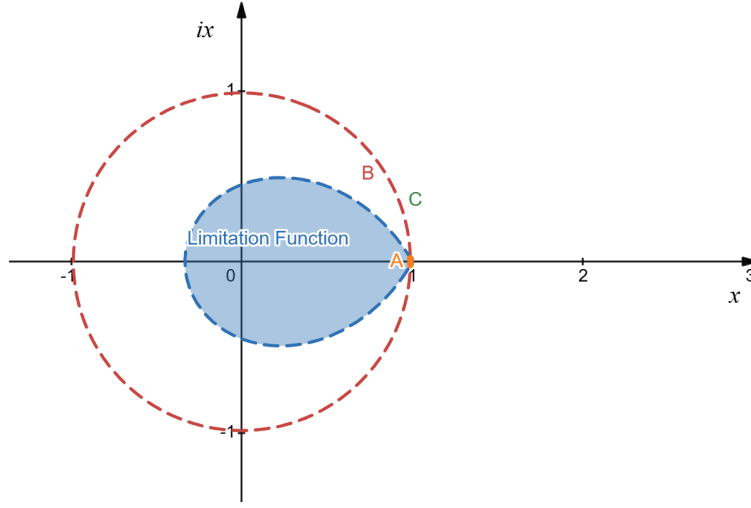


Figure 1: Curves A, B and C in the complex plane when $n = 10000$

Then consider $|z|^n = \left(1 - \frac{\pi}{\sqrt{6n}}\right)^n \sim \exp\left(-\frac{\pi\sqrt{n}}{\sqrt{6}}\right)$, $|1 - z| \sim \frac{\pi\sqrt{2}}{\sqrt{3n}}$, and $|z| \sim 1$, we have

$$\begin{aligned} LHS &= O\left(\int_B |z|^{-n} \left\{ \exp\left(\frac{\pi^2}{6|1-z|}\right) + \exp\left(|1-z|^{-1} + (1-|z|)^{-1}\right) \right\} |dz|\right) \\ &= O\left(\int_B \exp\left(\pi\sqrt{\frac{n}{6}}\right) \left\{ \exp\left(\frac{\pi}{6}\sqrt{\frac{3n}{2}}\right) + \exp\left(\frac{1}{\pi}\left(\sqrt{\frac{3n}{2}} + \sqrt{6n}\right)\right) \right\} |dz|\right) \\ &= O\left(\exp\left(\frac{\pi}{6} + \frac{1}{\pi}\left(\sqrt{\frac{3}{2}} + \sqrt{6}\right)\right) \sqrt{n}\right), \end{aligned}$$

where $\left(\pi/6 + 1/\pi\left(\sqrt{3/2} + \sqrt{6}\right)\right) \approx 1.693 < \pi\sqrt{2/3} \approx 2.565$. Then we turn to A. We apply Theorem 3.2 first to get

$$\begin{aligned} \int_A \frac{f(z) - \phi(z)}{z^{n+1}} dz &= O\left(\int_A \frac{|\phi(z)O(1-z)|}{|z|^n} dz\right) \\ &= O\left(\int_A |z|^{-n} |1-z|^{1/2} \exp\left(\frac{\pi^2}{6} \frac{1}{1-z}\right) \cdot |1-z| |dz|\right) \\ &= O\left(\int_A |z|^{-n} |1-z|^{3/2} \exp\left(\frac{\pi^2}{6} \frac{1}{1-|z|}\right) |dz|\right), \end{aligned}$$

which uses the estimate $|1-z| \sim 1-|z|$, since $1-|z| \leq |1-z| < 2(1-|z|)$. Then note that the length of A is $O(n^{-1/2})$, and for all $z \in C$, $|1-z| \sim 1-|z| = O(1/\sqrt{n})$, which means

$$\begin{aligned} LHS &= O\left(\left(\exp\left(2 \cdot \pi\sqrt{n/6}\right)\right) \cdot n^{-3/4}\right) \cdot O(n^{-1/2}) \\ &= O(n^{-5/4} \exp(\pi\sqrt{2n/3})). \end{aligned}$$

Finally, as the exponential term controls the whole integration, we conclude that

$$p(n) - q(n) = O\left(\int_C \frac{f(z) - \phi(z)}{z^{n+1}} dz\right) = O\left(n^{-\frac{5}{4}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right)\right).$$

Estimation of the Computable Function

We finally prove Theorem 3.4.

Theorem 3.4. *For large n , the partition function satisfies*

$$p(n) \sim q(n) \sim \frac{\exp\left(\pi\sqrt{2n/3}\right)}{4n\sqrt{3}}.$$

Proof. For $|z| < 1$, we use Theorem 2.8 (the Gaussian integral) to compute that

$$\begin{aligned} (1-z) \int_{-\infty}^{\infty} \exp\left[\pi t \sqrt{\frac{2}{3}} - (1-z)t^2\right] dt &= (1-z) \cdot \sqrt{\frac{\pi}{1-z}} \exp\left(\frac{\left(\pi\sqrt{2/3}\right)^2}{4(1-z)}\right) \\ &= \sqrt{\pi(1-z)} \exp\left(\frac{\pi^2}{6(1-z)}\right) \\ &= \pi\sqrt{2} e^{\pi^2/12} \phi(z). \end{aligned}$$

Next, we use Theorem 2.1 (the Taylor expansion) of exponents to get

$$\begin{aligned} \pi\sqrt{2} e^{\pi^2/12} \phi(z) &= (1-z) \int_{-\infty}^{\infty} \exp\left[\pi t \sqrt{\frac{2}{3}} - t^2 + zt^2\right] dt \\ &= (1-z) \int_{-\infty}^{\infty} \exp\left[\pi t \sqrt{\frac{2}{3}} - t^2\right] \sum_{k=0}^{\infty} \frac{z^k t^{2k}}{k!} dt \\ &= \int_{-\infty}^{\infty} \exp\left[\pi t \sqrt{\frac{2}{3}} - t^2\right] \sum_{k=0}^{\infty} \frac{z^k t^{2k}}{k!} dt - \int_{-\infty}^{\infty} \exp\left[\pi t \sqrt{\frac{2}{3}} - t^2\right] \sum_{k=0}^{\infty} \frac{z^{k+1} t^{2k+2}}{k!} dt. \end{aligned}$$

Comparing the coefficients, we have

$$\begin{aligned} \pi\sqrt{2} e^{\pi^2/12} q(n) &= \int_{-\infty}^{\infty} \exp\left(\pi\sqrt{\frac{2}{3}}t - t^2\right) \left(\frac{t^{2n}}{n!} - \frac{t^{2n-2}}{(n-1)!}\right) dt \\ &= \int_{-\infty}^{\infty} \exp\left(\pi\sqrt{\frac{2}{3}}t - t^2\right) \frac{t^{2n}}{n!} \left(1 - \frac{n}{t^2}\right) dt. \end{aligned}$$

With Theorem 2.9 (Stirling's approximation formula) and the substitution $t = \sqrt{n} + s$, there exists

$$\begin{aligned}
LHS &= \int_{-\infty}^{\infty} \exp\left(\pi\sqrt{\frac{2}{3}}(s + \sqrt{n}) - s^2 - 2\sqrt{n}s - n\right) \frac{(s + \sqrt{n})^{2n}}{n^n e^{-n} \sqrt{2\pi n}} \left(\frac{s^2 + s\sqrt{n}s}{s^2 + 2\sqrt{n}s + n}\right) ds \\
&= k \int_{-\infty}^{\infty} \exp\left(\pi\sqrt{\frac{2}{3}}s - s^2 - 2\sqrt{n}s\right) \left(\frac{s + \sqrt{n}}{\sqrt{n}}\right)^{2n} \left(\frac{s^2 + s\sqrt{n}s}{s^2 + 2\sqrt{n}s + n}\right) ds \\
&= k \int_{-\infty}^{\infty} \exp\left(\pi\sqrt{\frac{2}{3}}s - s^2 - 2\sqrt{n}s\right) \left(1 + \frac{s}{\sqrt{n}}\right)^{2n-2} \frac{(s^2 + 2\sqrt{n}s + n)(s^2 + 2\sqrt{n}s)}{n(s^2 + 2\sqrt{n}s + n)} ds \\
&= k \int_{-\infty}^{\infty} \exp\left(\pi\sqrt{\frac{2}{3}}s - s^2 - 2\sqrt{n}s\right) \left(1 + \frac{s}{\sqrt{n}}\right)^{2n-2} \frac{s^2 + 2\sqrt{n}s}{n} ds,
\end{aligned}$$

where we set $k = \exp\left(\pi\sqrt{2n/3}\right) / \sqrt{2\pi n}$ for simplicity. Then extracting $\frac{1}{\sqrt{n}}$ and s from the last part gives that

$$LHS = \frac{\exp\left(\pi\sqrt{\frac{2n}{3}}\right)}{\sqrt{2\pi n}} \int_{-\infty}^{\infty} s \exp\left(\pi\sqrt{\frac{2}{3}}s - s^2 - 2\sqrt{n}s\right) \left(1 + \frac{s}{\sqrt{n}}\right)^{2n-2} \left(2 + \frac{s}{\sqrt{n}}\right) ds.$$

Subsequently, we eliminate n in the integration with respect to s by using limits. We use the function

$$F(s) = \begin{cases} s \exp\left(\pi\sqrt{\frac{2}{3}}s - s^2\right) (2 + s) & (s \geq 0), \\ |s| \exp\left(\pi\sqrt{\frac{2}{3}}s - s^2\right) (2 - s) \exp(s^2 + 1) & (s < 0), \end{cases}$$

in Theorem 2.10 (the dominated convergence theorem) to make sure that the limit sign and the integration sign can exchange positions. To meet the requirements in DCT, we now discuss some claims of the functions that we use.

Claim 3.4.1. For all $s \in \mathbb{R}$,

$$\left| s \exp\left(\pi\sqrt{\frac{2}{3}}s - s^2 - 2\sqrt{n}s\right) \left(1 + \frac{s}{\sqrt{n}}\right)^{2n-2} \left(2 + \frac{s}{\sqrt{n}}\right) \right| \leq F(s).$$

Proof. We set

$$A(s) = e^{-2\sqrt{n}s} \left(1 + \frac{s}{\sqrt{n}}\right)^{2n-2} \left(2 + \frac{s}{\sqrt{n}}\right),$$

while

$$\begin{aligned}
B_1(s) &= 2 + s, \\
B_2(s) &= (2 - s) \exp(s^2 + 1).
\end{aligned}$$

Now, it is clear that we can just compare $A(s)$ with $B_1(s)$ and $B_2(s)$.

1. $s \geq 0$. We have $A(0) = B_1(0) = 2$, while $B_1'(s) > 0$ but $A'(s) < 0$ and $A(s) > 0$, which means that $|A(s)| \leq B_1(s)$.

2. $s < 0$. We want to show

$$|A(s)| = e^{-2\sqrt{n}s} \left(1 + \frac{s}{\sqrt{n}}\right)^{2n-2} \left|2 + \frac{s}{\sqrt{n}}\right| \leq B_2(s) = (2-s) \exp(s^2 + 1),$$

or

$$\left(1 + \frac{s}{\sqrt{n}}\right)^{2n-2} \frac{\left|2 + \frac{s}{\sqrt{n}}\right|}{2-s} \leq \exp(s^2 + 2s\sqrt{n} + 1).$$

To prove this, note that for $n \geq 1$, there is

$$\frac{\left|2 + \frac{s}{\sqrt{n}}\right|}{2-s} \leq 1$$

for all $s < 0$, so

$$\left(1 + \frac{s}{\sqrt{n}}\right)^{2n-2} \frac{\left|2 + \frac{s}{\sqrt{n}}\right|}{2-s} \leq \left(1 + \frac{s}{\sqrt{n}}\right)^{2n-2} \leq \exp\left(\frac{s}{\sqrt{n}}(2n-2)\right).$$

Finally, as $n \geq 1$, we have

$$s^2 + 2s\sqrt{n} + 1 - \frac{s}{\sqrt{n}}(2n-2) = \left(s - \frac{1}{\sqrt{n}}\right)^2 + 1 - \frac{1}{n} > 0.$$

Therefore, we complete our proof. □

Claim 3.4.2. $\int_{-\infty}^{\infty} F(s)ds$ converges for all $n \in \mathbb{N}^*$.

Proof. We separate the integral into two parts.

1. For $s \geq 0$, the integral can be divided into two parts, each using Corollary 2.11 (the derivatives of Gaussian integral).
2. For $s < 0$, the integral is convergent and computable from integration by parts:

$$\begin{aligned} \int_{-\infty}^0 F(s)ds &= \int_{-\infty}^0 -s \exp\left(\pi\sqrt{\frac{2}{3}}s - s^2\right) (2-s) \exp(s^2 + 1) ds \\ &= \int_{-\infty}^0 -s \exp\left(\pi\sqrt{\frac{2}{3}}s + 1\right) (2-s) ds \\ &= \frac{3e}{\pi^2} + \frac{3\sqrt{6}e}{2\pi^3}. \end{aligned}$$

□

These claims ensure that DCT works, so now we discuss the result of using the limit in the integral sign. We show that

$$\lim_{n \rightarrow \infty} e^{-2\sqrt{n}s} \left(1 + \frac{s}{\sqrt{n}}\right)^{2n-2} \left(2 + \frac{s}{\sqrt{n}}\right) = 2e^{-s^2},$$

based on Theorem 2.1 (the Taylor expansion):

$$\left(1 + \frac{s}{\sqrt{n}}\right)^{2n-2} = \exp \left[(2n-2) \ln \left(1 + \frac{s}{\sqrt{n}}\right) \right] \approx \exp \left[(2n-2) \left(\frac{s}{\sqrt{n}} - \frac{s^2}{2n} \right) \right] \approx \exp(2\sqrt{n}s - s^2).$$

Finally, we can use Theorem 2.10 (DCT) and Theorem 2.8 (the Gaussian integral) to deduce that

$$\begin{aligned} \pi\sqrt{2}e^{\pi^2/12}q(n) &\sim \frac{\exp\left(\pi\sqrt{\frac{2n}{3}}\right)}{\sqrt{2\pi n}} \int_{-\infty}^{\infty} s \exp\left(\pi\sqrt{\frac{2}{3}}s - s^2 - 2\sqrt{n}s\right) \left(1 + \frac{s}{\sqrt{n}}\right)^{2n-2} \left(2 + \frac{s}{\sqrt{n}}\right) ds \\ &\sim \frac{\exp\left(\pi\sqrt{\frac{2n}{3}}\right)}{\sqrt{2\pi n}} \int_{-\infty}^{\infty} s \exp\left(\pi\sqrt{\frac{2}{3}}s - s^2\right) \cdot 2e^{-s^2} ds. \\ &= \frac{\pi}{2\sqrt{6n}} \exp\left(\pi^2/12 + \pi\sqrt{2n/3}\right), \end{aligned}$$

or

$$q(n) \sim \frac{\exp\left(\pi\sqrt{2n/3}\right)}{4n\sqrt{3}}.$$

Finally, note that from Theorem 3.3, $p(n) - q(n) = O\left(n^{-5/4} \exp\left(\pi\sqrt{2n/3}\right)\right)$. Since the error term grows slower than the main term, we complete the proof. $\square$

4 Further Development on Estimation

Ramanujan's partition formula has inspired further research on the partition function. In 1918, Hardy and Ramanujan not only provided Ramanujan's partition formula, a basic estimate, but also gave a full estimate [7]:

$$p(n) = \frac{1}{2\pi\sqrt{2}} \sum_{k=1}^N A_k(n) \frac{d}{dn} \left[\frac{\sinh\left(\frac{\pi}{k}\sqrt{\frac{2}{3}\left(n - \frac{1}{24}\right)}\right)}{\sqrt{n - \frac{1}{24}}} \right] + O(n^{-1/4}),$$

where $A_k(n)$ are Kloosterman-type sums and N is the number of terms included in the finite sum [2].

The final major step was carried out by Rademacher in 1937, who proved a convergent series expansion by extending the circle method, expressed as an infinite series involving Kloosterman-type sums and Bessel functions [2, 16]:

$$p(n) = \frac{1}{\pi\sqrt{2}} \sum_{k=1}^{\infty} A_k(n) \sqrt{k} \frac{d}{dn} \left(\frac{\sinh\left(\frac{\pi}{k}\sqrt{\frac{2}{3}\left(n - \frac{1}{24}\right)}\right)}{\sqrt{n - \frac{1}{24}}} \right).$$

This formula marks the culmination of efforts to express the partition function in explicit numerical

terms.

5 Significance of Ramanujan's Partition Formula

Building on these refinements and extensions, we now turn to the broader significance of Ramanujan's partition formula.

5.1 Ramanujan's Partition Formula in Approximation

Before the derivation of Ramanujan's partition formula, calculating $p(n)$ for large n was computationally intractable. Direct enumeration—even for $n \geq 200$ —fails due to the rapid growth of $p(n)$, and early recurrence relations such as Euler's pentagonal number theorem require storing all prior values of $p(n)$, which is inefficient for large-scale computations [17]. In this background, Ramanujan's partition formula provided one of the most efficient and accurate estimates.

Ramanujan's partition formula has two key advantages:

1. The error term grows significantly slower than the main term. For example, there exists $p(100) = 190569292$, while the formula's estimate is approximately 199280893, with a relative error of less than 4.6%; for $n = 500$, the relative error drops to roughly 2% (Figure 3 in the appendix section).
2. The formula only requires basic exponential and square-root operations. This efficiency makes it a critical tool for preliminary checks in number theory, especially for verifying the order of magnitude of large numbers.

5.2 Connections to Restricted Partitions

Along with normal partitions, people also focus on some special partitions. Take the restricted partition function only with odd parts, $p_{\text{odd}}(n)$, as an example. Its generating function is:

$$f_{\text{odd}}(z) = \sum_{n=0}^{\infty} p_{\text{odd}}(n)z^n = \prod_{k=1}^{\infty} \frac{1}{1 - z^{2k-1}} \quad (|z| < 1),$$

which is a variant of the basic partition generating function $f(z)$. By modifying the order estimate of $|f(z)|$ and applying the circle method (as in the proof of Ramanujan's partition formula), an analogous asymptotic formula for $p_{\text{odd}}(n)$ is derived [7]:

$$p_{\text{odd}}(n) \sim \frac{\exp\left(\pi\sqrt{\frac{n}{3}}\right)}{2n\sqrt{3}}.$$

This result mirrors Ramanujan's original formula but with a scaled exponent, reflecting the smaller number of restricted partitions (e.g., $p_{\text{odd}}(4) = 3$ vs. $p(4) = 5$). Such adaptations confirm that Ramanujan's methodological framework, rather than just the formula itself, generalizes to constrained partition problems.

5.3 Connections to Other Mathematical Fields

Using tools introduced in Section 2 and incorporating knowledge in other mathematical fields, Ramanujan’s partition formula is not isolated to partition theory; it connects to foundational areas of mathematics. We discuss two of the related fields here.

5.3.1 Complex Analysis

The proof of Ramanujan’s partition formula in this article relies entirely on complex analysis, as introduced in Section 3. The key steps—using the Cauchy’s integral formula to extract $p(n)$ from the generating function in terms of complex integrals (Theorem 2.6), applying Cauchy’s theorem to simplify contour integrals (Theorem 2.5), and estimating complex integrals via order analysis (Theorem 3.1)—demonstrate how complex analysis enables solving combinatorial problems. This cross-field connection shows that even elementary combinatorial objects (like partitions) require advanced analytic tools to understand their asymptotic behavior, bridging discrete and continuous mathematics.

5.3.2 Modular Forms and q-Series

The generating function of partitions, though appearing as a purely combinatorial object at first glance, has deep connections to complex analysis and number theory. Specifically, the generating function of partitions can be interpreted as the reciprocal of the Dedekind eta function

$$\eta(\tau) = q^{1/24} \prod_{k=1}^{\infty} (1 - q^k),$$

which is a modular form of weight $1/2$ [3]. Modular forms are complex functions defined on the upper half-plane that exhibit precise symmetry under some special rules. This symmetry allows one to control the behavior of $f(q)$ near singularities, ultimately leading to Ramanujan’s partition formula. Readers may learn about modular forms from textbooks[3].

In other words, the combinatorial problem of counting integer partitions is naturally linked to analytic structures with special symmetry properties, and this connection provides a powerful method for deriving precise asymptotic estimates.

6 Conclusion

In this article, we provide a detailed proof of Ramanujan’s partition formula for the partition function using the circle method and complex analysis, in a manner accessible to readers with a high-school level background in mathematics. By carefully explaining each step and the underlying ideas, we make this classic result more comprehensible. We also highlighted the significance of Ramanujan’s partition formula in the study of integer partitions and its connections to other areas of mathematics. However, our exposition focuses on the classical case and provides only an introductory-level treatment, so some advanced aspects, generalizations, and deeper analytic techniques are not covered. Thus, future work may consider further generalizations of the partition

function, refined asymptotic estimates, and investigations into restricted partitions as well as other combinatorial problems.

References

- [1] George E. Andrews. *The Elementary Theory of Partitions*, page 1–15. Encyclopedia of Mathematics and its Applications. Cambridge University Press, 1984. No DOI available.
- [2] George E. Andrews. *The Hardy–Ramanujan–Rademacher Expansion of $p(n)$* , page 68–87. Encyclopedia of Mathematics and its Applications. Cambridge University Press, 1984. No DOI available.
- [3] Krishnaswami Alladi and Frank Garvan. *Partitions, q -series, and modular forms*, volume 23. Springer Science & Business Media, 2011. No DOI available.
- [4] Chi-Chun Zhou and Wu-Sheng Dai. A statistical mechanical approach to restricted integer partition functions. *Journal of Statistical Mechanics: Theory and Experiment*, 2018(5): 053111, May 2018. doi: <https://doi.org/10.1088/1742-5468/aabfc9>.
- [5] George E. Andrews. *Infinite Series Generating Functions*, page 16–32. Encyclopedia of Mathematics and its Applications. Cambridge University Press, 1984. No DOI available.
- [6] George E. Andrews. *Restricted Partitions and Permutations*, page 33–53. Encyclopedia of Mathematics and its Applications. Cambridge University Press, 1984. No DOI available.
- [7] G. H. Hardy and S. Ramanujan. Asymptotic formulæ in combinatory analysis. *Proceedings of the London Mathematical Society*, s2-17(1):75–115, 1918. doi: <https://doi.org/10.1112/plms/s2-17.1.75>.
- [8] James V Uspensky. Asymptotic formulae for numerical functions which occur in the theory of partitions. *Bulletin of the Russian Academy of Sciences*, 14(6):199–218, 1920. No DOI available.
- [9] D. J. Newman. The evaluation of the constant in the formula for the number of partitions of n . *American Journal of Mathematics*, 73(3):599–601, 1951. doi: <https://doi.org/10.2307/2372311>.
- [10] D. J. Newman. A simplified proof of the partition formula. *Michigan Math. J.*, 9(1):283–287, 1962. doi: <https://doi.org/10.1307/mmj/1028998729>.
- [11] James Stewart. *Calculus: Early Transcendentals*. Brooks/Cole, Cengage Learning, Belmont, CA, 7th edition, 2012. ISBN 9780538497909. No DOI available.
- [12] Alessio Del Vigna. On a solution to the basel problem based on the fundamental theorem of calculus, 2021.
- [13] E. Saff, A. Snider, L.N. Trefethen, and T.A. Driscoll. *Fundamentals of Complex Analysis with Applications to Engineering and Science (Classic Version)*. Pearson Modern Classics for Advanced Mathematics Series. Pearson Education, 2017. ISBN 9780134689487. URL <https://books.google.com.hk/books?id=-5MFMQAACAAJ>. No DOI available.

- [14] George Pólya and Gabor Szegő. *Problems and theorems in analysis I*. Springer Study Edition. Springer, Berlin, Germany, 1976. No doi available.
- [15] Timothy D. Browning. *The Hardy—Littlewood circle method*, pages 123–149. Birkhäuser Basel, Basel, 2009. ISBN 978-3-0346-0129-0. doi: https://doi.org/10.1007/978-3-0346-0129-0_8.
- [16] Hans Rademacher. On the partition function $p(n)$. *Proceedings of the London Mathematical Society*, s2-43(1):241–254, 1938. doi: <https://doi.org/10.1112/plms/s2-43.4.241>.
- [17] Béla Bollobás. *Recurrence Relations for the Partition Function*, page 326–327. Cambridge University Press, 2022.

Appendix

Numerical Accuracy of Ramanujan's Partition Formula

It is visible that the formula gives an accurate estimate of $p(n)$, and the relative error decreases dramatically as $n \rightarrow \infty$ (Figure 2 and Figure 3).

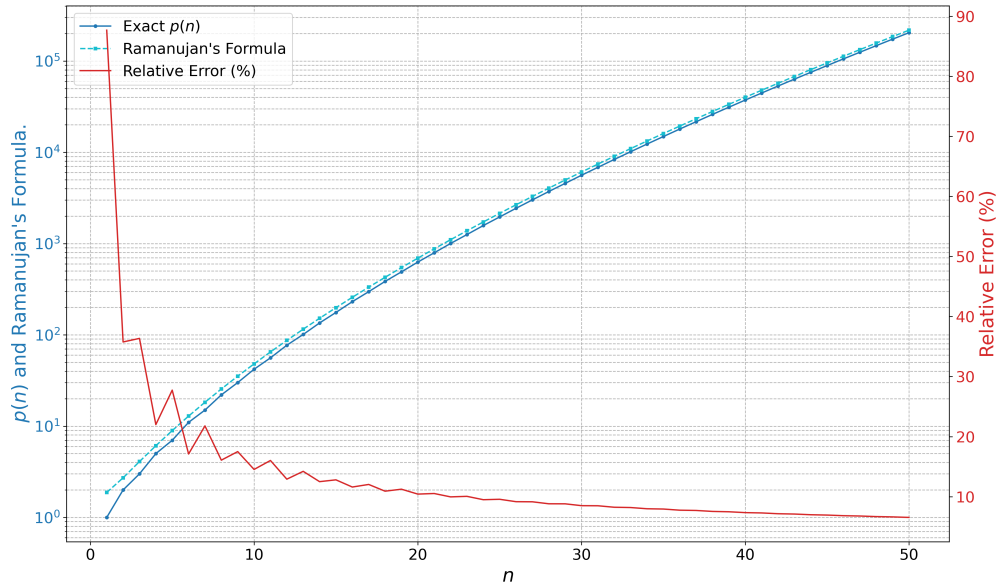


Figure 2: Partition Function and Ramanujan's Partition Formula for $n \leq 50$

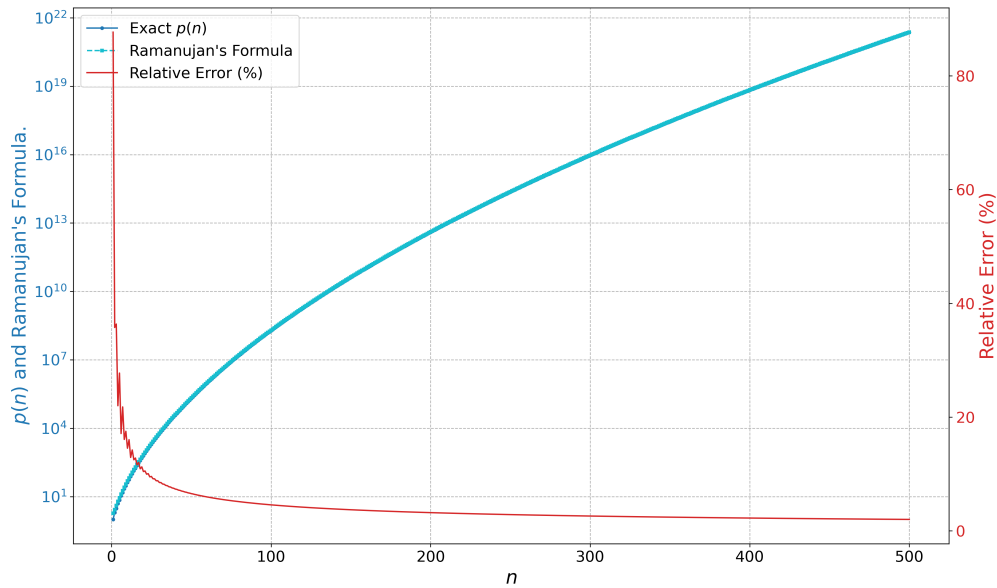


Figure 3: Partition Function and Ramanujan's Partition Formula for $n \leq 500$